

CSE5DMI Data Mining

Week 01

La Trobe University CRICOS Provider Code Number 00115M

Teaching Team

Lecturer

- Dr Choiru (c.zain@latrobe.edu.au)

- Research interest
  - Large-scale Data Stream Analytics
  - Machine Learning & Artificial Intelligence
  - Big Data Analytics
  - Data Mining

- LinkedIn: https://www.linkedin.com/in/choiru-za-in-452358167/

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Teaching Team

Subject Coordinator

- Dr Lydia Cui (L.Cui@latrobe.edu.au)
- Senior Lecturer, CSIT
- CSIT Department Teaching & Learning Coordinator (2023 - now)
- MICT/MIT/MAI course Coordinator (2022-2025)
- Background
  - PhD in computer science, The University of Sydney
  - Postdoctoral research fellow, The University of Sydney
  - Research area: image processing, medical computer vision, machine learning
  - Webpage: https://scholars.latrobe.edu.au/display/lcui

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Teaching Team

Lab Demonstrator

Jianqiao Zhang



- Final year PhD student
- Bachelor with Honours graduate in 2022
- Research focus: applying artificial intelligence in the medical field, particularly in early disease prediction using medical images, clinical data, and radar-based medical tasks.
- "Outside of studies, I am a guitarist and photographer. I enjoy rock, blues, jazz, and funk music, and I used to play in a band. Additionally, I am a member of the La Trobe University Society of Music (LUSOM). I also have a passion for vintage lenses and cameras.

I hope you all have a great new semester, and I look forward to seeing you in class!"

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Objectives

This subject

- Provides students experience and knowledge in the emerging field of data mining

Upon successful completion of this subject, you should be able to:

1. Assess the quality and suitability of data for mining tasks and develop effective data cleaning, reduction and transformation solutions.
2. Analyse a given data science problem and recommend data mining strategies suitable to solve the problem.
3. Implement classification, clustering, and association rule mining techniques for pattern discovery from selected datasets.
4. Evaluate the performance of data mining models using appropriate metrics.
5. Present data-driven insights by written reports and visualizations and make data-informed recommendations to technical and non-technical audiences.

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LEARNING ACTIVITIES

Session	Learning Focus and Activities	Contact hours
1	Lecture – Introduction to Data Mining . No labs.	2
2	Lecture – Data and Similarity; Lab – Python Basics Review	4
3	Lecture – Ruled-based Classification Techniques (1); Lab – Python Data Analysis Library	4
4	Lecture – Rule-based Classification Techniques (2); Lab – Data pre-processing	4
5	Lecture – Learning-based Classification Techniques (1); Lab – Data visualisation	4
6	Lecture – Learning-based Classification Techniques (2); Lab – Decision Tree	4
7	Lecture – Basics of Association Analysis; Lab – Assignment Consultation	4
8	Lecture –Association Analysis; Lab – Neural Network Classification	4
9	Lecture –Rule-clustering Techniques (1); Lab – Support Vector Machine	4
10	Lecture –Rule-clustering Techniques (2); Lab – K-means Clustering	4
11	Lecture –Case study – Industry application; Lab – Hierarchical Clustering	4
12	Lecture – Review; Lab – Assignment Consultation	4
	Total	46

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Study Mode & Weekly activities

Lecture (One 2-hour face2face lecture per week)

- Tuesday 11:00 - 13:00
- Lectures are automatically recorded by Echo 360

Echo360

Weekly consultation:

- *please email me to make an appointment.*

It is very important to come to labs. Practice Makes Perfect!

Practice (one 2-hour face2face lab per week from Week 2)

- Labs are not recorded nor delivered online.
- If you cannot attend labs regularly, there will be high risks of failing the assignments. Please re-consider enrolling in the subject.
- Join your allocated session according to Allocate+
  - *Wed 9:00 - 11:00, PS2-123, Jianqiao*
  - *Wed 11:00-13:00, PS2-123. Yijie*
  - *Wed 13:00-15:00, BG108. Jianqiao*

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Study Mode & Weekly activities

Post-class quiz (one quiz per week)

- There will be ten (10) tests throughout the semester, with the first test starting in week one of the semester.
- Weighting value: 10% (1% per week in designed teaching weeks; 10×1%= 10% in total)
- Details of task: Format: online homework, multiple-choice, problem-solving.
- Each test will only be available through LMS for one week, that is from Tuesday 1 pm in the topic week to Tuesday 11 am next week.
- You will have two attempts to do the quiz in each week before the quiz is closed.
- The questions in the two attempts can be different. The higher grade will be recorded.

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Assessments

Week/ Session	%	Assessment type	Key Criteria	Feedback method	SILOs assessed	Eligible for short extension?
Weekly	10	Completion of 10 weekly online tests	Completion of online quizzes and evaluated by staff.	Online via LMS.	SILO 1,3,5	No
Midnight, Sunday, Week TBC	20	Individual Assignment. Students engage with predefined datasets to perform comprehensive data exploration and analysis.	TBA in assignment description	Online written comments by teaching staff	SILO1, SILO5	Yes
Midnight, Sunday, Week TBC	30	Individual Programming assignment. Source code and a written report on knowledge discovery problems from a given dataset.	TBA in assignment description	Online written comments by teaching staff	SILO1, SILO2, SILO3, SILO4, SILO5	Yes
Central Exam Period	40	2-hour **closed book face to face (paper based)** examination.		Final Results	SILO1, SILO2, SILO4, SILO5	No

The subject cannot grant any students with online exams. Please re-consider whether this is the subject you want to enrol.

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Get familiar with LTU Policies

- All coursework students are eligible for Short Extension of Time.
- You will automatically receive a Short Extension of Time of 3 calendar days, and students with a Learning Access Plan (LAP) will receive an extension of 7 calendar days upon the provision of their LAP. Please see below for more information for students with a LAP.
- If you require a further extension, beyond the Short Extension of Time, you should apply for Special Consideration (if eligible).
- **Request for Short Extensions of time to submit an assessment task**
  https://www.latrobe.edu.au/students/admin/forms/request-an-extension
  - Applications for Short Extensions must be submitted by all students, using the Adjustments Form, no later than 11.59pm (Melbourne time) on the due date of the assessment.
  - Students without a LAP are not required to submit supporting evidence with the Short Extension application. Late applications for Short Extensions are not accepted from students who do not have a LAP.
  - Students with a LAP that includes extensions to submission dates are required to provide a copy of their LAP as part of any application for an extension that is greater than three days.

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Get familiar with LTU Policies

- **Special consideration https://www.latrobe.edu.au/students/admin/forms/special-consideration**
  - To be considered eligible for Special Consideration, you must be able to prove you have experienced circumstances considered to be of short-term impact, outside of your control that have or will have a significant impact on your ability to complete an assessment task.
  - Once you have identified that your circumstance may may align with the requirements for Special Consideration, you must:
    - Provide the necessary documentation outlined in Reasons and Documentation to support your application
    - Submit your application before the due date of your assessment
  - Eligibility matrix: https://www.latrobe.edu.au/students/admin/forms/assessment-adjustments/eligibility-criteria

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Get familiar with LTU Policies

- **Late submission penalties**
  - Five per cent of the total possible mark for the task (ie 5% of 100%) per calendar day that the task is overdue.
  - Work will not be assessed if the submission is overdue by more than five (5) calendar days after:
    - the original submission date without an approved extension; or
    - any approved extension for that student (including those granted under any Learning Access Plan or Special Consideration application).
- **Academic integrity**
  - The penalty for submitting an assignment under your name that is the work of a third party may be severe, even leading to exclusion from the University without readmission. Refer to the **Academic Integrity - Schedule of Penalties and Actions** within the Student Academic Misconduct Policy

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CSE5DMI Web Page - LMS

Announcement

Subject information

Weekly learning materials

- Lecture note
- Lecture recording
- Lab note
- Post-class quiz

Assignments

University policies

- Special consideration
- Plagiarism
- etc

Students are expected to check the LMS and announcements regularly

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Contact us

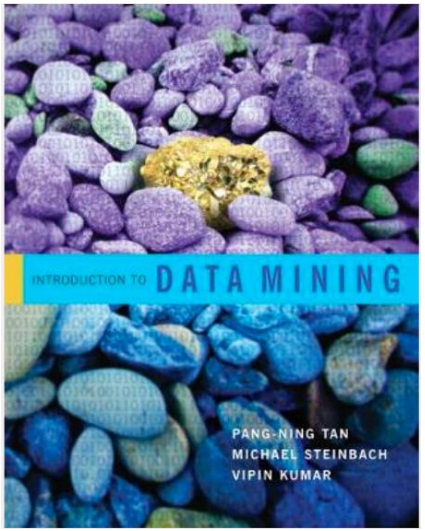
- Our class is very big.
- We receive huge number of emails from students enrolled in different subjects everyday.

We would appreciate it if you could email us using your *LTU account* and emphasize *CSE5DMI* in the email subject line.

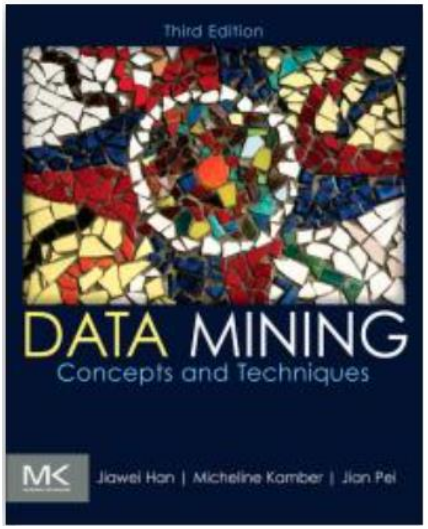
- For any questions about lectures, please contact Choiru.
- Questions about lab contents, contact lab demonstrators, and copy (cc) Choiru in your request
- For assessment extension & special consideration applications, please contact Lydia, and copy (cc) Choiru in your email.

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Reference book



- **Introduction to Data Mining,** Book by Michael Steinbach, Pang-Ning Tan, and Vipin Kumar
  - Selected chapters available via https://www-users.cse.umn.edu/~kumar001/dmbook/index.php
- **Data Mining: Concepts and Techniques : Concepts and Techniques**, Book by Jiawei Han, Micheline Kamber, and Jian Pei
  - Available via LTU library https://ebookcentral-proquest-com.ez.library.latrobe.edu.au/lib/latrobe/detail.action?docID=729031



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Large-scale Data is Everywhere!

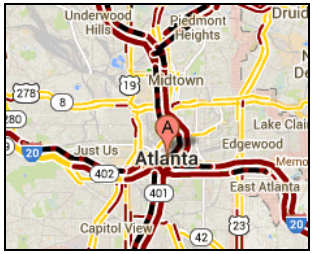
- There has been enormous data growth in both commercial and scientific databases due to advances in data generation and collection technologies
- New mantra
  - Gather whatever data you can whenever and wherever possible.
- Expectations
  - Gathered data will have value either for the purpose collected or for a purpose not envisioned.



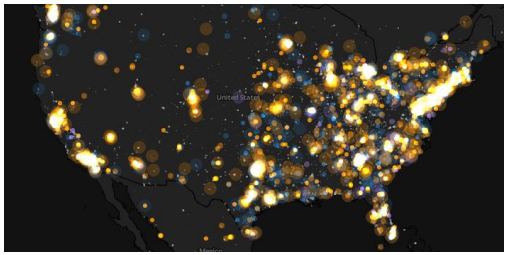
Cyber Security



E-Commerce



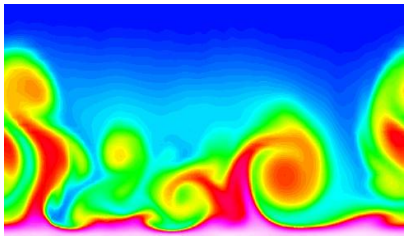
Traffic Patterns



Social Networking: Twitter



Sensor Networks



Computational Simulations

Why Mine Data? Commercial Viewpoint

- Lots of data is being collected and warehoused
  - Web data, e-commerce
  - purchases at department/ grocery stores
  - Bank/Credit Card transactions







- Computers have become cheaper and more powerful
- Competitive Pressure is Strong
  - Provide better, customized services for an *edge* (e.g. in Customer Relationship Management)

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Why Mine Data? Scientific Viewpoint



Cyber Security

Network traffic
Login records
Firewalls
Servers
User activities

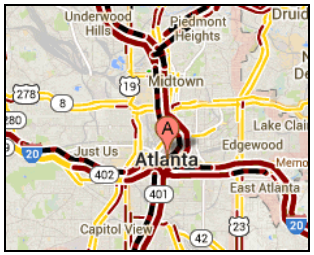
A data-mining system learns what normal network behaviour looks like



E-Commerce

customer searches,
page views,
purchases,
reviews, and
shopping-cart activity

Recommender system with association rule (e.g. Laptop + laptop bag → wireless mouse)



Traffic Patterns

Number of vehicles per minute
Average vehicle speed
Location
Weather conditions
Time and day
Roadworks, etc

Traffic on a particular road becomes heavily congested between 7:30 am and 9:00 am on weekdays, especially when it rains.

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Why Mine Data? Scientific Viewpoint

From data to pattern

Cyber security: learn normal behaviour and flag abnormal activity.
E-commerce: discover product combinations and recommend related items.
Traffic: identify congestion patterns by time, weather, and road condition.

Example: traffic congestion rule

Input features:
- Vehicle count per minute
- Average speed
- Weather
- Time and day
- Roadworks

Possible pattern:
“Road A is likely congested between 7:30–9:00 on weekdays, especially during rain.”

Key message: Data mining helps convert repeated observations into actionable patterns.

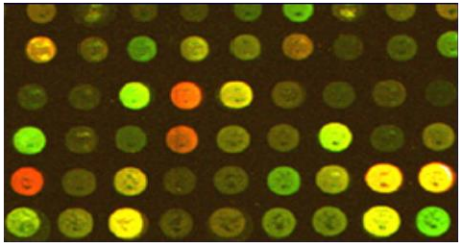
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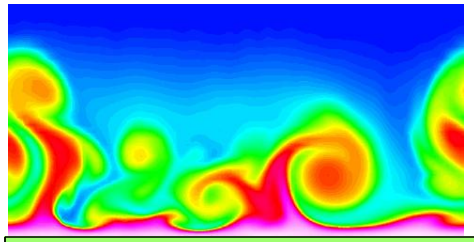
Why Mine Data? Scientific Viewpoint

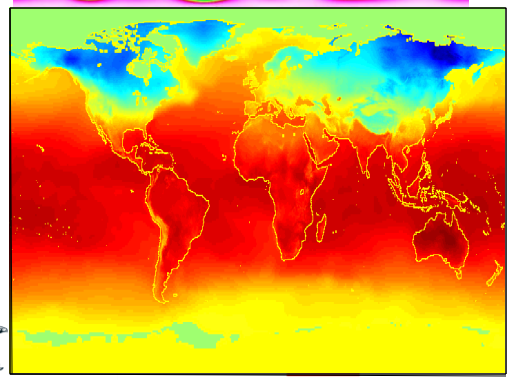
- Data collected and stored at enormous speeds (GB/hour)
  - remote sensors on a satellite
  - telescopes scanning the skies
  - microarrays generating gene expression data
  - scientific simulations generating terabytes of data
- Traditional techniques infeasible for raw data
- Data mining may help scientists
  - in classifying and segmenting data
  - in Hypothesis Formation







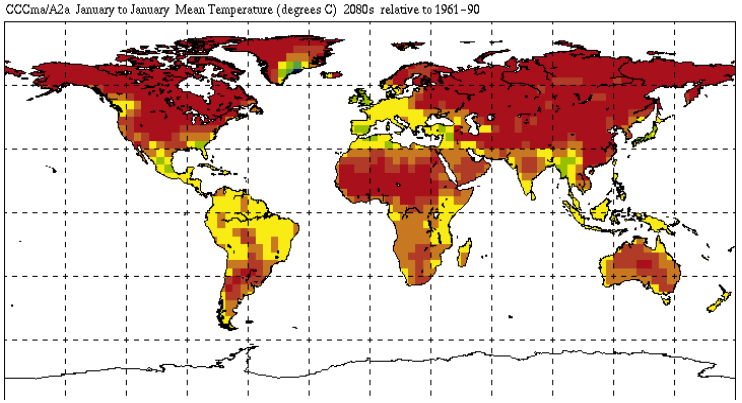




Great Opportunities to Solve Society's Major Problems



Improving health care and reducing costs



Predicting the impact of climate change



Finding alternative/ green energy sources



Reducing hunger and poverty by increasing agriculture production

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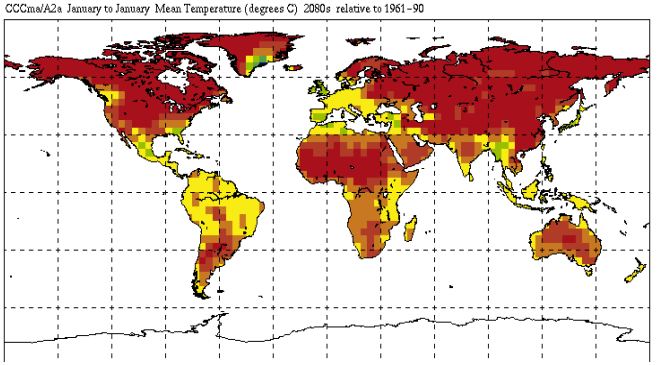
How Data Mining Solve The Problems?



1. Healthcare: Improving Healthcare and Reducing Costs

- **Features:** Age, blood pressure, heart rate, medical history, lab results, length of stay.
- **Prediction:** Will the patient be readmitted within 30 days?

Benefits: Early intervention, reduced hospital costs, improved patient care.



2. Climate Change: Predicting Climate Impact

- **Features:** Temperature, rainfall, CO_2 level, sea level, humidity, forest coverage.
- **Prediction:** Drought, flood, or wildfire risk in the coming years.
- **Benefits:** Better disaster preparedness, resource planning, and environmental protection.

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Quiz: A farmer wants to predict next season's crop

1. Which of the following features would you collect?

- **Soil nutrient level**
- **Rainfall**
- **Number of follower in social media follower**
- **Temperature**
- **Type of fertilizer**
- **Frequency of Irrigation**
- **Farmer favourite footy's team**

2. What should model predict?

- **The farmer's income**
- **Crop output next season**
- **Number of tractors sold**
- **Tomorrow's weather prediction**

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How Data Mining Solve The Problems?



3. Renewable Energy: Finding Green Energy Sources

- **Features:** Solar radiation, wind speed, rainfall, land price, distance to power grid.
- **Prediction:** Best location for a solar or wind farm.
- **Benefits:** Higher energy efficiency, lower investment risk, sustainable energy production.

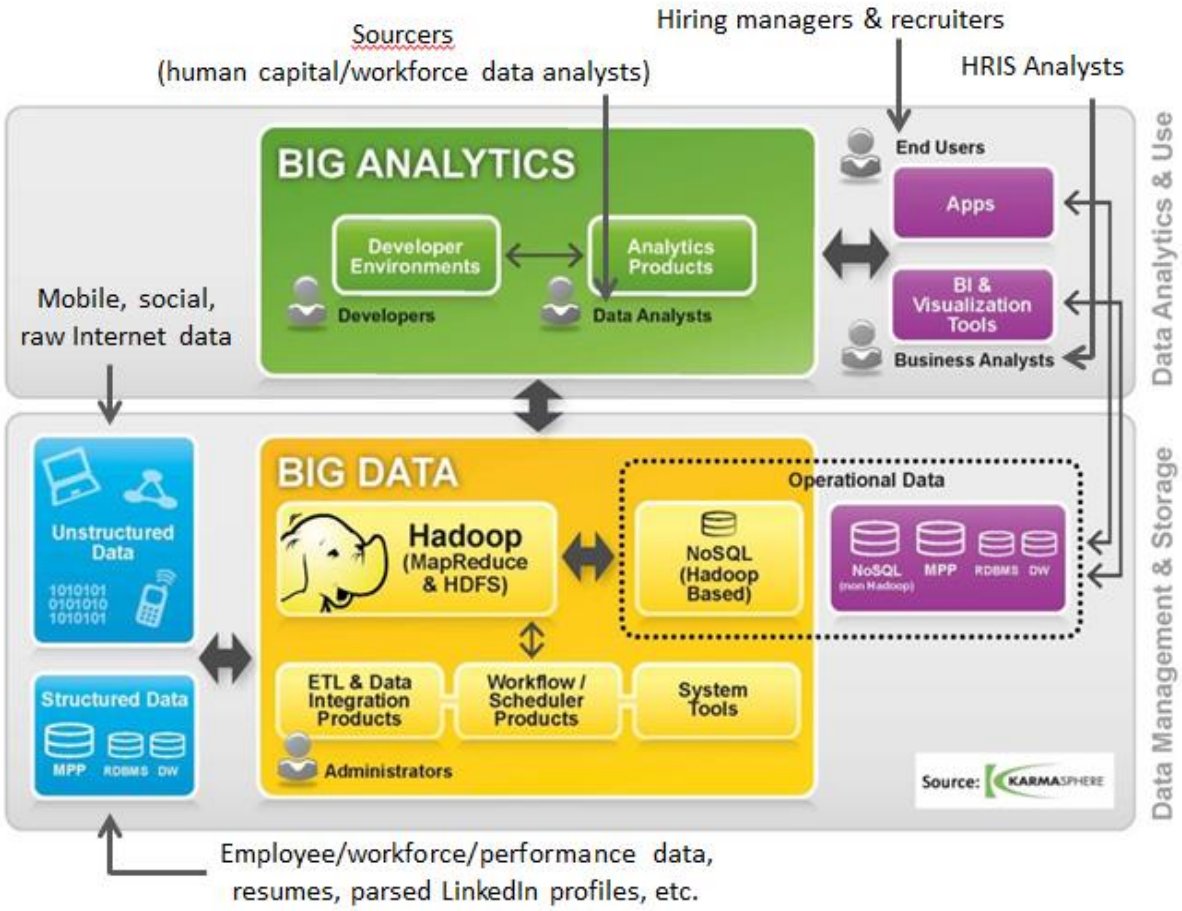


4. Agriculture: Increasing Crop Production

- **Features:** Soil nutrients, rainfall, temperature, fertilizer usage, irrigation, pest attacks.
- **Prediction:** Crop yield and disease risk for the next season.
- **Benefits:** Higher productivity, reduced costs, improved food security.

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Some applications – Human resources

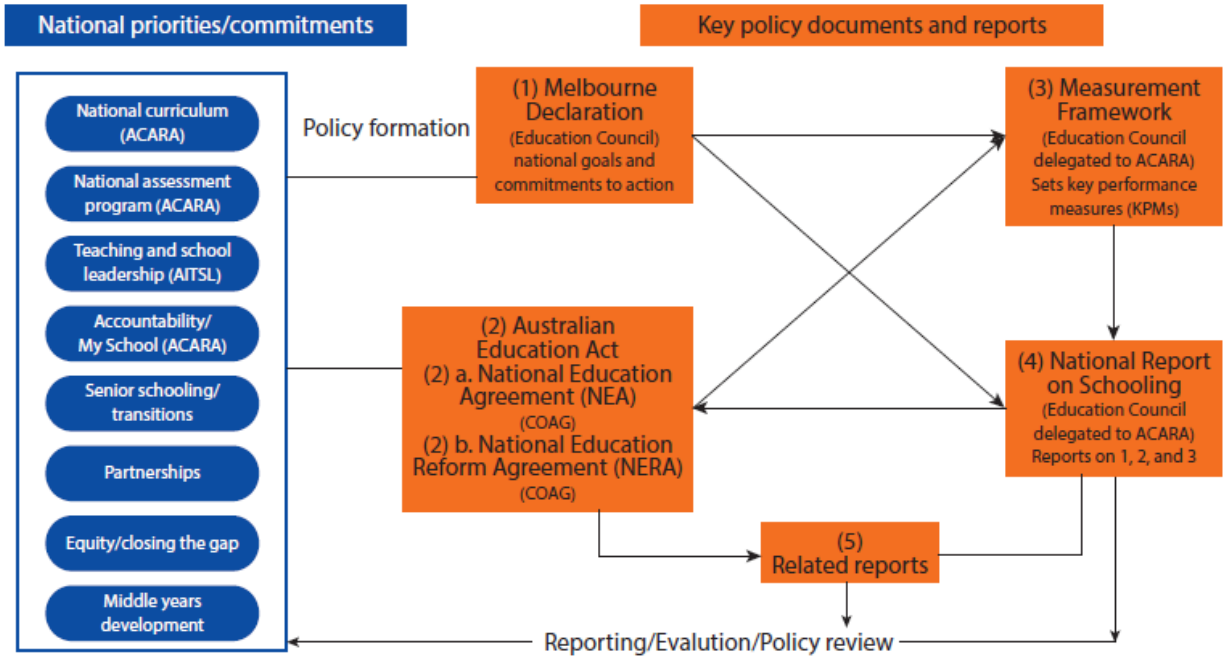


Big data / big analytics graphic of various types of human capital data. ©Karmasphere

Big data / big analytics graphic from Karmasphere showing examples of the types of relevant human capital data (structured and unstructured) that can be leveraged by various domain/functional experts – sourcers, recruiters, HRIS analysts, and hiring managers.

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Some applications - Education



Improving the transparency and accountability of educational data

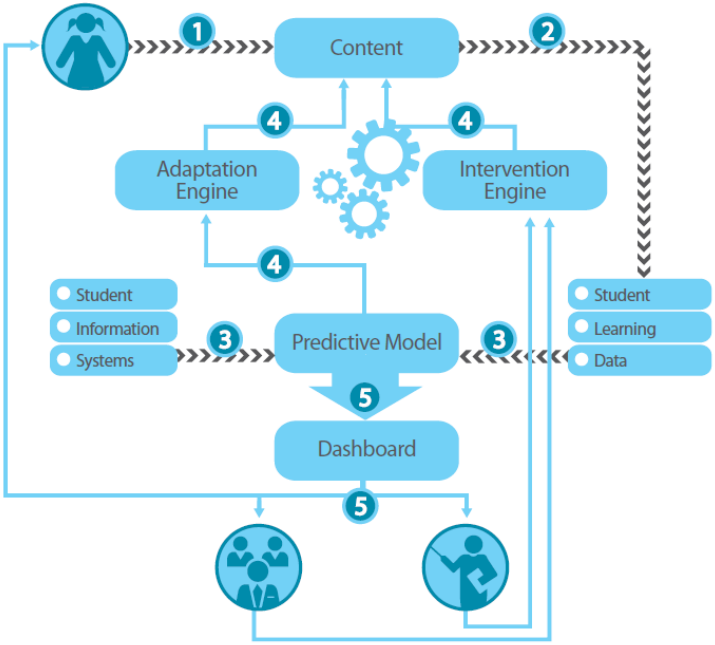
Australian National School Education Policy Framework

"Schools Assistance Act 2008" and "Australian Education Act 2013“

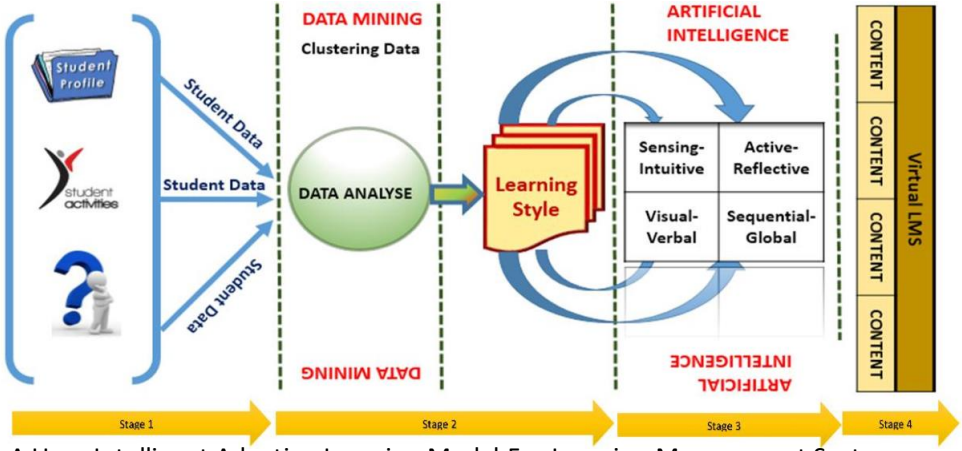
- Assist national agencies in developing national standards and collecting quantitative data
- Annual school board report
- Covers: learning and teaching, student environment, leadership and management, community participation

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Some applications - Education



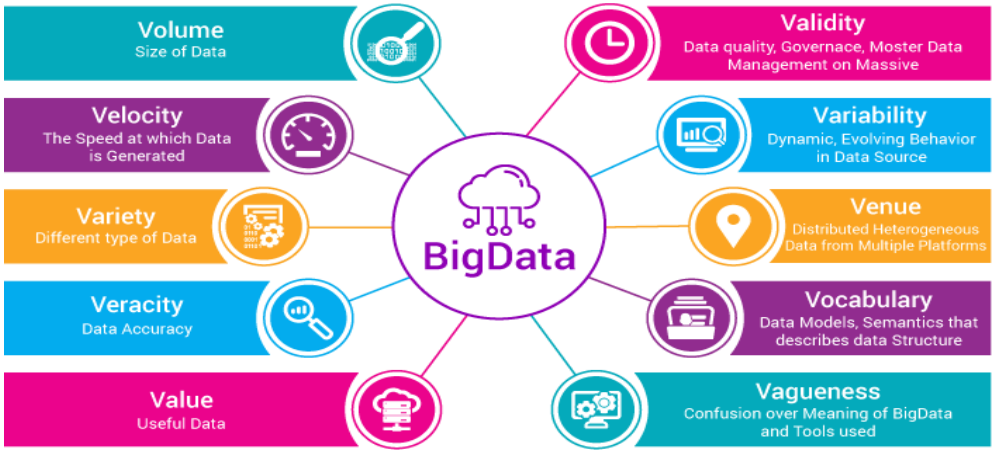
© Dreambox, adaptive learning model

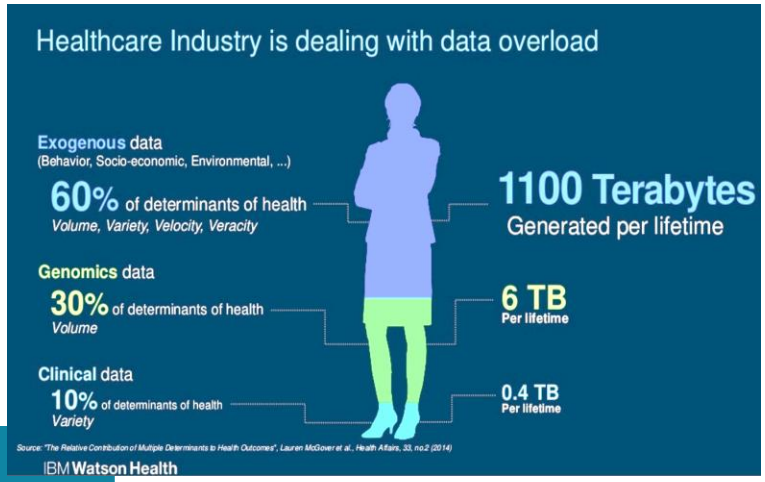


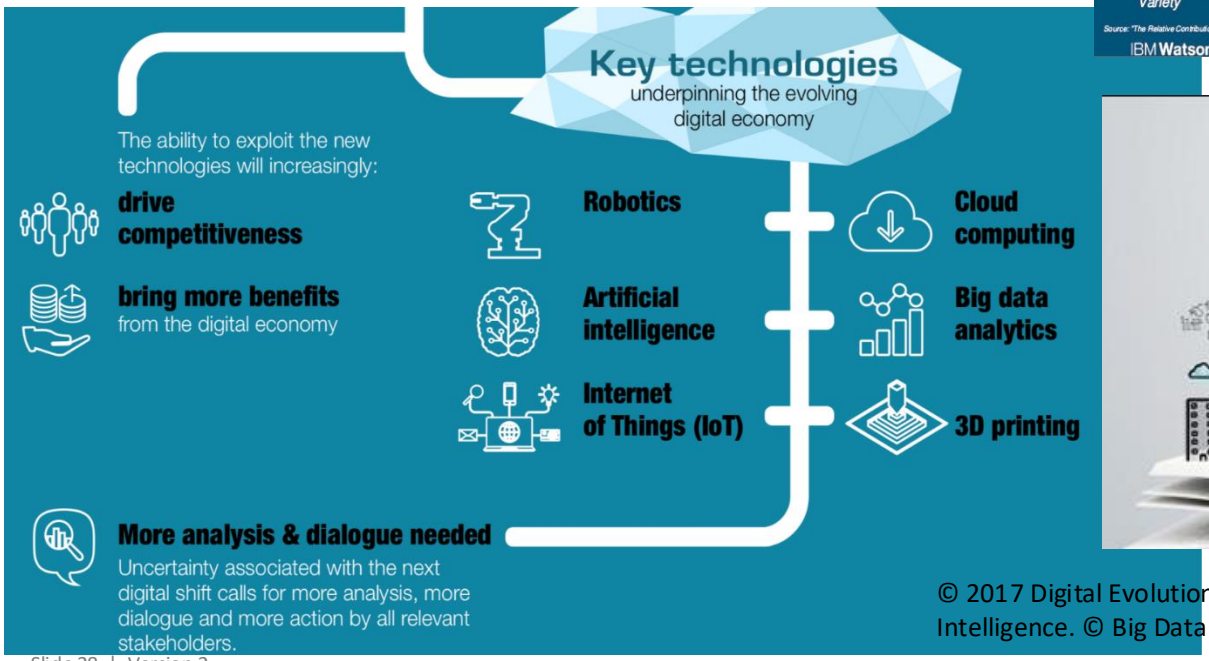
A User-Intelligent Adaptive Learning Model For Learning Management System Using Data Mining And Artificial Intelligence

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Some applications - Healthcare









© 2017 Digital Evolution Report – CyberCrime, Digitization, Blockchain and Artificial Intelligence. © Big Data Hadoop Pro. © IBM Watson Health

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AI + Big Data in Biomedicine

Prediction

- Lab results for disease onset prediction

Prediction of the onset of diseases by analyzing patient electronic health records (EHRs)

EHRs

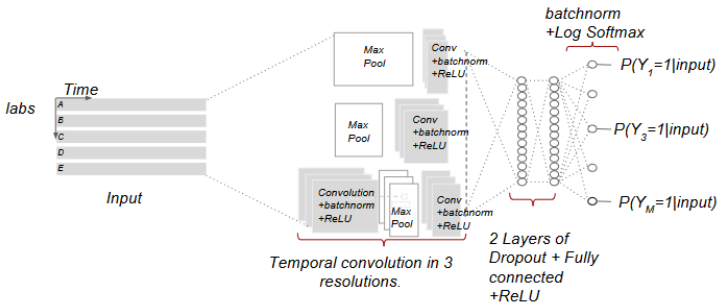
Lab tests

X-rays

.....

Doctor's notes

- 133 conditions
- 18 common lab tests
- 298K patients
- 8 years administrative claims data



Prediction

Severe kidney disease

Hormone-related conditions

Heart failure

Liver problems

Diabetes

...

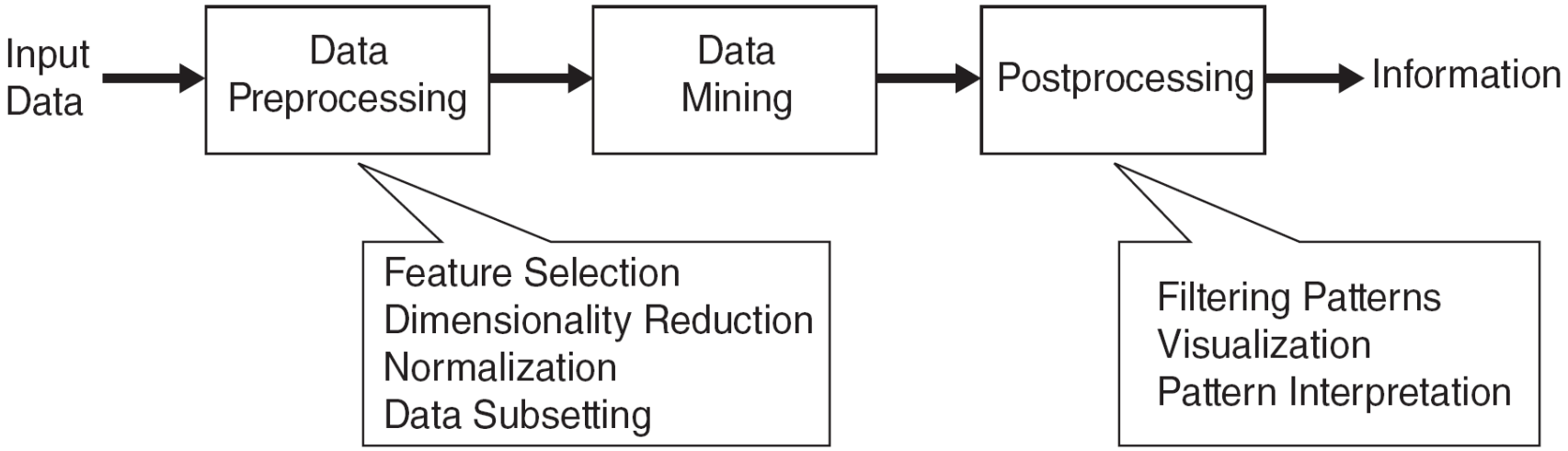
three months earlier than traditional methods [1]

[1] Narges Razavian, Jake Marcus, David Sontag,"Multi-task Prediction of Disease Onsets from Longitudinal Lab Tests", Machine Learning and Healthcare, 2016

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What is Data Mining?

- Many Definitions
  - Non-trivial extraction of implicit, previously unknown and potentially useful information from data
  - Exploration & analysis, by automatic or semi-automatic means, of large quantities of data in order to discover meaningful patterns



What is (not) Data Mining?

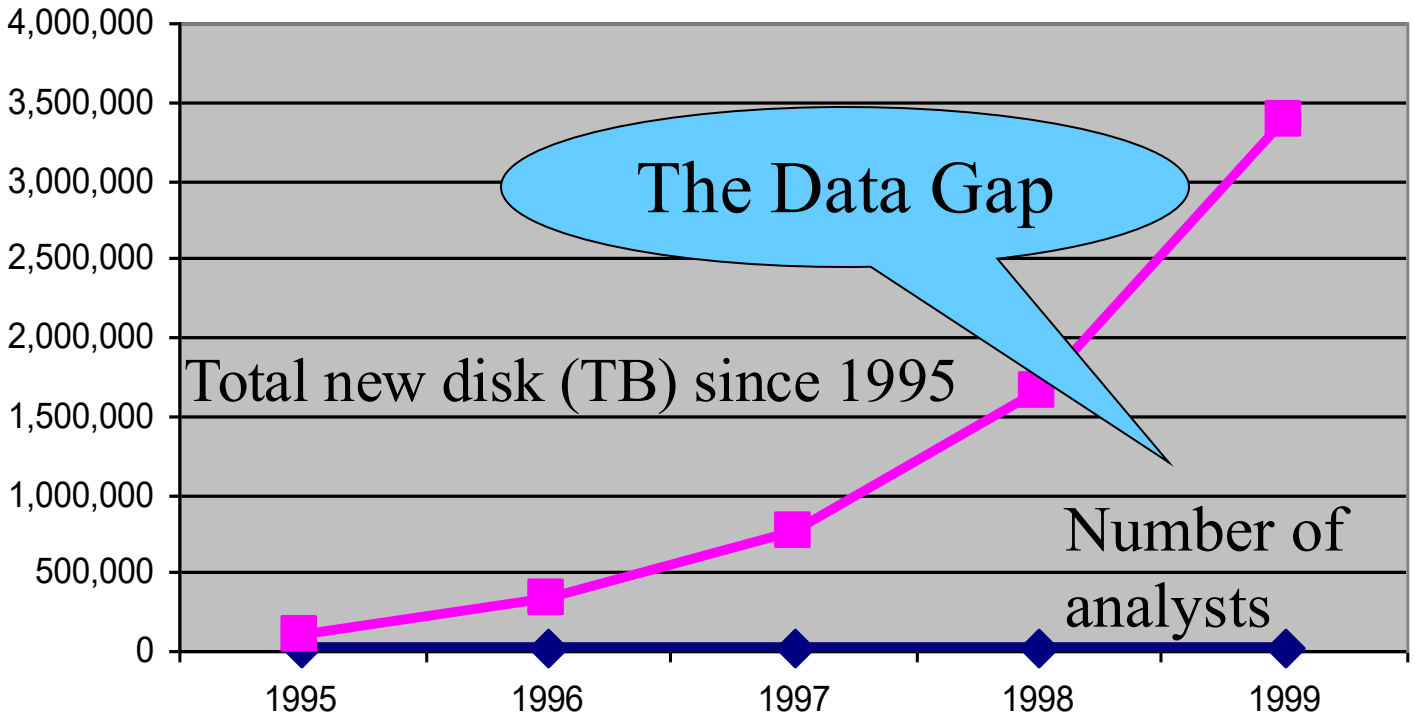
- What is not Data Mining?
  - Look up phone number in phone directory
  - Query a Web search engine for information about "Amazon"

- What is Data Mining?
  - Certain names are more prevalent in certain US locations (O'Brien, O'Rurke, O'Reilly… in Boston area)
  - Group together similar documents returned by search engine according to their context (e.g. Amazon rainforest, Amazon.com,)

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Mining Large Data Sets - Motivation

- There is often information "hidden" in the data that is not readily evident
- Human analysts may take weeks to discover useful information
- Much of the data is never analyzed at all

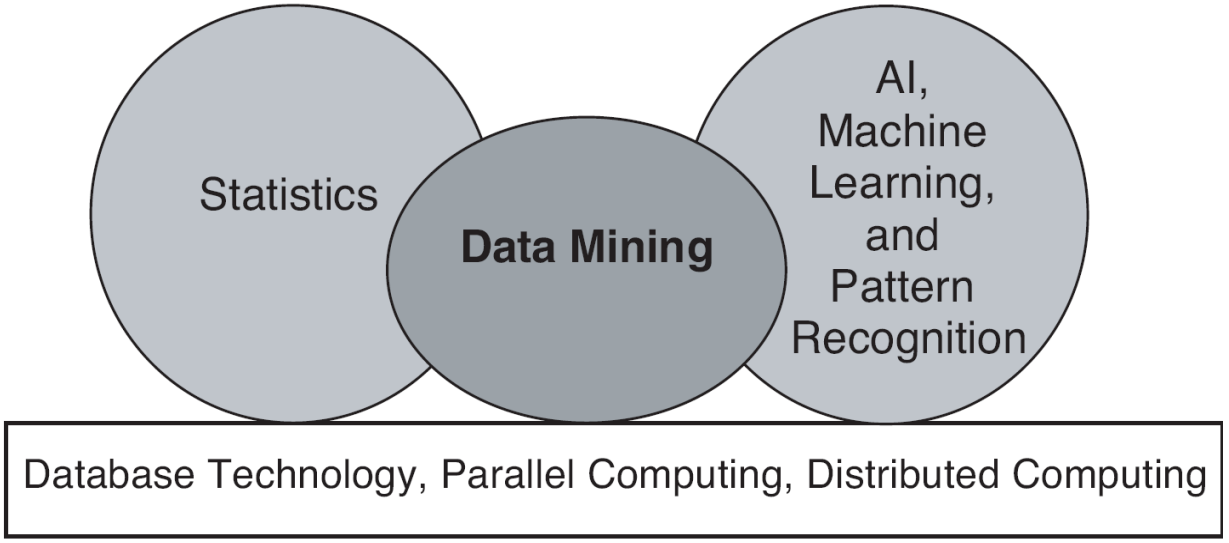


From: R. Grossman, C. Kamath, V. Kumar, "Data Mining for Scientific and Engineering Applications"

Origins of Data Mining

- Draws ideas from machine learning/AI, pattern recognition, statistics, and database systems

- Traditional techniques may be unsuitable due to data that is
  - Large-scale
  - High dimensional
  - Heterogeneous
  - Complex
  - Distributed

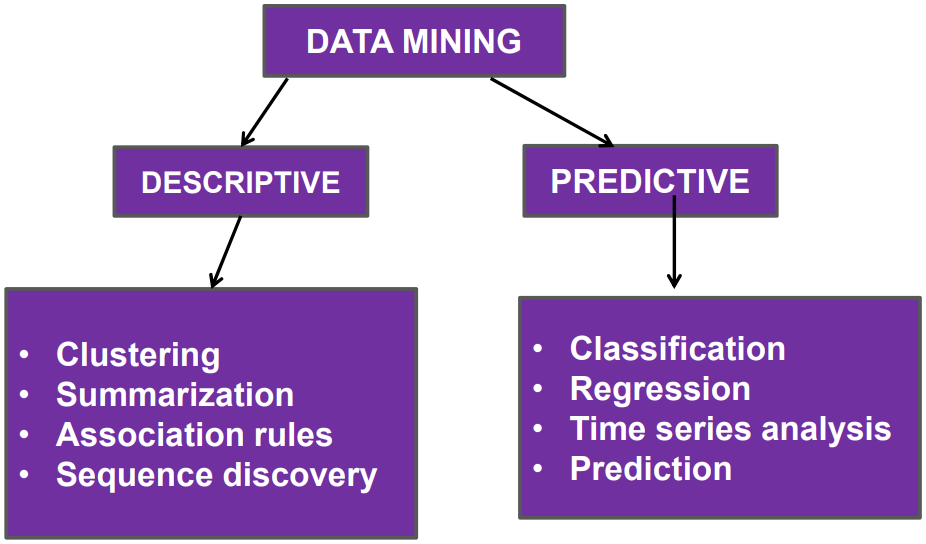


- A key component of the emerging field of data science and data-driven discovery

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Data Mining Tasks

- Prediction Methods
  - Use some variables to predict unknown or future values of other variables.
- Description Methods
  - Find human-interpretable patterns that describe the data.



From [Fayyad, et.al.] Advances in Knowledge Discovery and Data Mining, 1996

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Quiz – Prediction or Description?

- Prediction Tasks
  - Classification; Predict Category
  - Regression; Predict a Numerical Value
  - Time Series Analysis; Forecast Future values
- Descriptive Tasks
  - Clustering; group similar object
  - Association Rules; finding relationships (e.g. bread -> butter)
  - Sequence Discovery; find ordered patterns

- Prediction Tasks
  - Which customers have similar shopping habits? (Prediction or Description)
  - Will a patient develop diabetes? (Prediction or Description)
  - How much the price of a particular house in the next decade? (Prediction or Description)
  - Will this email be spam? (Prediction or Description)
  - Will a customer buy this product? (Prediction or Description)
  - Which products are frequently purchased together? (Prediction or Description)
  - What are the common characteristics of high-performing students? (Prediction or Description)

From [Fayyad, et.al.] Advances in Knowledge Discovery and Data Mining, 1996

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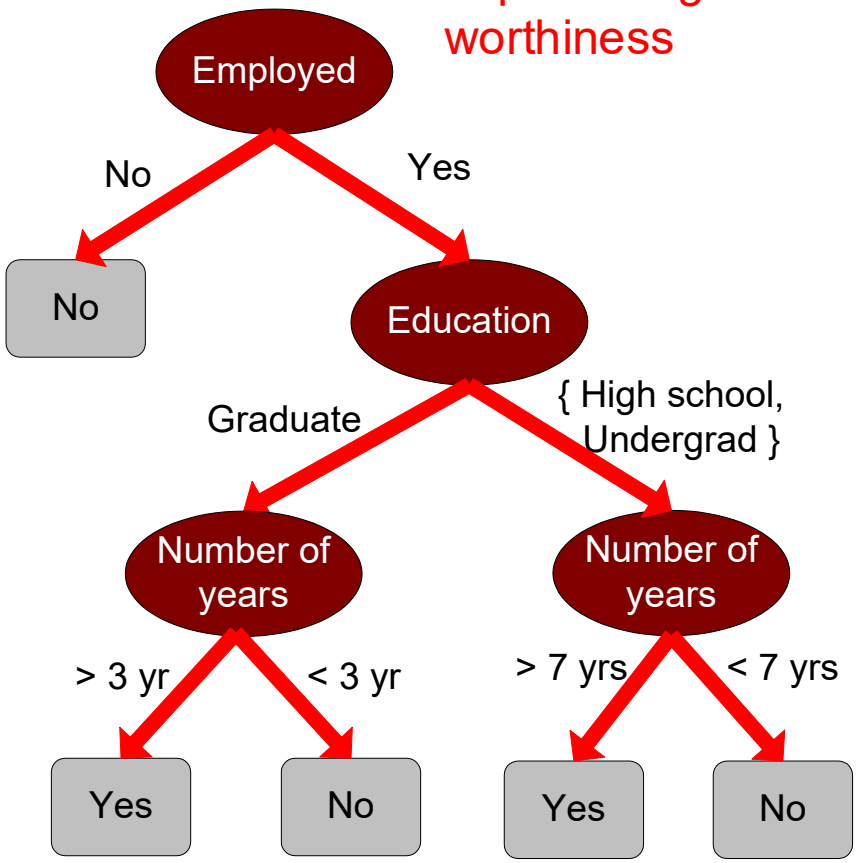
Predictive Modeling: Classification

- Given a collection of records (training set), each record contains a set of attributes, one of the attributes is the class.
- Find a model for class attribute as a function of the values of other attributes
- Goal: previously unseen records should be assigned a class as accurately as possible.

Class

Tid	Employed	Level of Education	# years at present address	Credit Worthy
1	Yes	Graduate	5	Yes
2	Yes	High School	2	No
3	No	Undergrad	1	No
4	Yes	High School	10	Yes
…	…	…	…	…

Model for predicting credit worthiness



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Examples of Classification Task

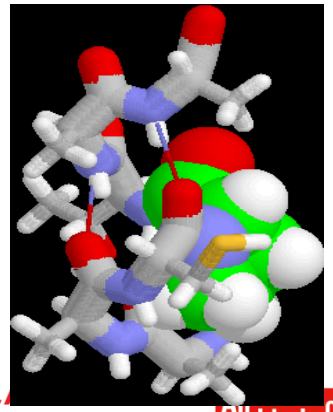
- Classifying credit card transactions as legitimate or fraudulent



- Classifying land covers (water bodies, urban areas, forests, etc.) using satellite data



- Categorizing news stories as finance, weather, entertainment, sports, etc
- Identifying intruders in the cyberspace
- Predicting tumor cells as benign or malignant



- Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil

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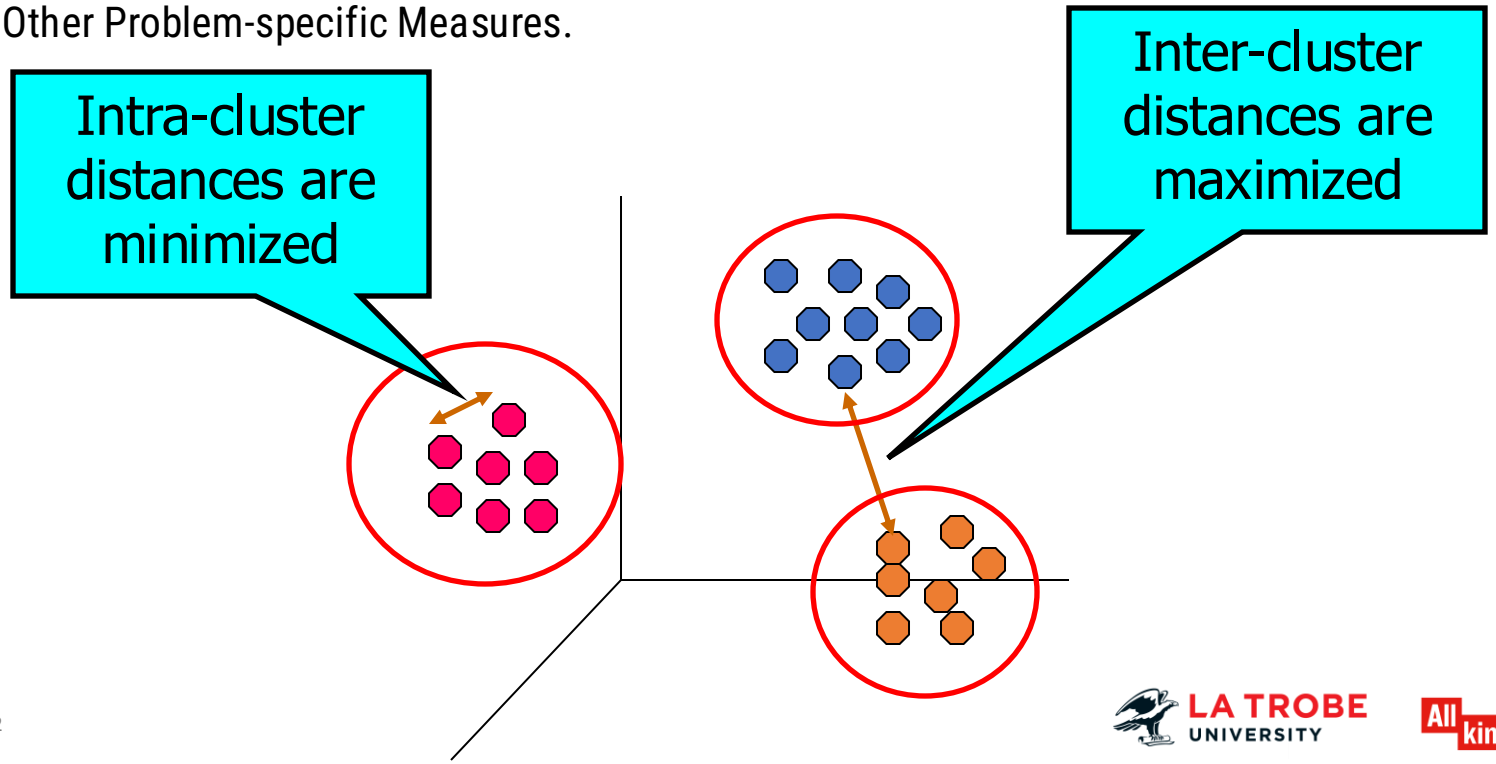
Regression

- Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.
- Extensively studied in statistics, neural network fields.
- Examples:
  - Predicting sales amounts of new product based on advetising expenditure.
  - Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
  - Time series prediction of stock market indices.

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Clustering

- Given a set of data points, each having a set of attributes, and a similarity measure among them, find clusters such that
  - Data points in one cluster are more similar to one another.
  - Data points in separate clusters are less similar to one another.
- Similarity Measures:
  - • Euclidean Distance if attributes are continuous.
  - • Other Problem-specific Measures.



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Clustering: Application

Market Segmentation:

- **Goal:** subdivide a market into distinct subsets of customers where any subset may conceivably be selected as a market target to be reached with a distinct marketing mix.
- **Approach:**
  - Collect different attributes of customers based on their geographical and lifestyle related information.
  - Find clusters of similar customers.
  - Measure the clustering quality by observing buying patterns of customers in same cluster vs. those from different clusters.

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Clustering: Application

Document Clustering:

- **Goal:** To find groups of documents that are similar to each other based on the important terms appearing in them.
- **Approach:** To identify frequently occurring terms in each document. Form a similarity measure based on the frequencies of different terms. Use it to cluster.
- Gain: Information retrieval can untilize the clusters to relate a new document or search them to clustered documents

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Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection
  - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}

Association Analysis: Applications

Marketing and Sales Promotion:

- Let the rule discovered be

  {Bagels, … } --> {Potato Chips}

- Potato Chips as consequent => Can be used to determine what should be done to boost its sales.
- Bagels in the antecedent => Can be used to see which products would be affected if the store discontinues selling bagels.
- Bagels in antecedent and Potato chips in consequent => Can be used to see what products should be sold with Bagels to promote sale of Potato chips!

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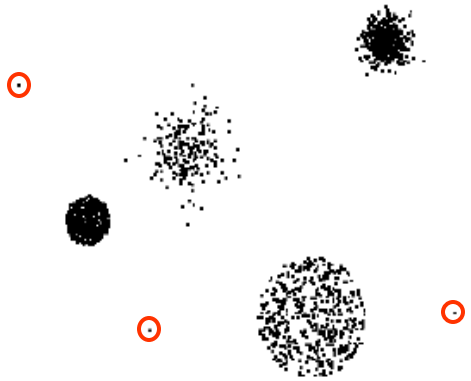
Association Analysis: Applications

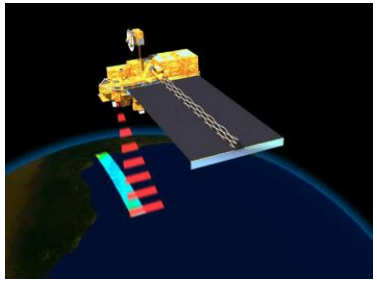
Supermarket shelf management:

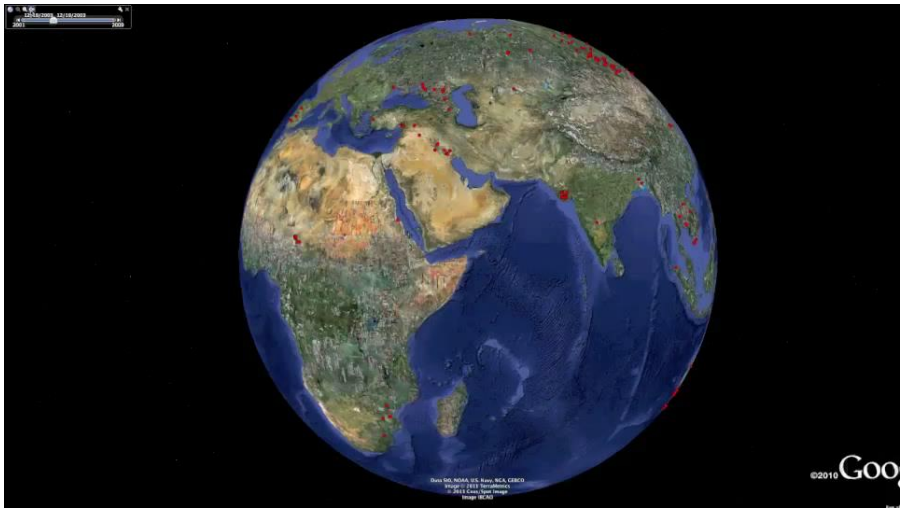
- Goal: To identify items that are bought together by sufficiently many customers.
- Approach: Process the point-of-sale data collected with barcode scanners to find dependencies among items.
- A classic rule -- e.g. {Diaper, Milk} --> {Beer}
  - If a customer buys nappy and milk, then he is very likely to buy beer.
  - So, don't be surprised if you find six-packs stacked next to nappies!

Deviation/Anomaly/Change Detection

- Detect significant deviations from normal behavior
- Applications:
  - Credit Card Fraud Detection
  - Network Intrusion Detection
  - Identify anomalous behavior from sensor networks for monitoring and surveillance.
  - Detecting changes in the global forest cover.







Motivating Challenges

- Scalability
- High Dimensionality
- Heterogeneous and Complex Data
- Data Ownership and Distribution
- Non-traditional Analysis

Summary

- Introduction to CSE5DMI
  - Weekly activities
  - Teaching team
  - Assessments
  - Important policies
- Introduction to data mining
  - Why data mining
  - What is data mining
  - Data mining tasks

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After class

Lecture 1 quiz will be available from 1pm today

- Go to LMS - > Week 1: Introduction to Data Mining - > POST-CLASS QUIZ

Get prepared for Lab 1 in Week 2

- Go to LMS - > Week 2: Data - > PRACTICAL LAB

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Thank you

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